

Total No. of Printed Pages:1

SUBJECT CODE NO: - RNP-8061
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Civil Engg.) (NEP) Sem - IV
Examination May/June 2026
Theory of Structure

[Time: 1:30 Hours]**[Max. Marks: 30]**

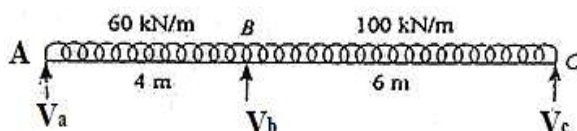
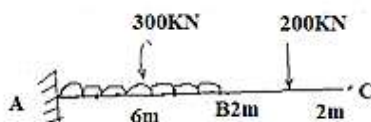
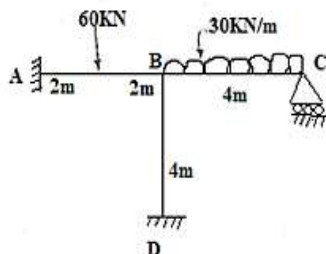
Please check whether you have got the right question paper.

N. B

1. Part A: Question No 1 is compulsory.
2. Attempt any Two questions from remaining Part B: Q.2 to Q.7.

Q.1 Solve any 5 questions from the following**10**

- i) Define degree of kinematic indeterminacy.
- ii) What is the principle of superposition?
- iii) Define carry-over factor in moment distribution method.
- iv) What is influence line diagram for shear force?
- v) Define conjugate beam method.
- vi) What is a two-hinged arch?
- vii) State assumptions in slope deflection method.
- viii) Define equivalent point load for moving loads.

Solve any TWO.**Q.2 Analyse the given beam by three moment method.****10****Q.3 Analyse the given beam by Slope Deflection Method.****10****Q.4 Analyse the given frame by using Moment Distribution Method.****10****Q.5 A three hinged parabolic arch has a span of 40m and a Central rise 6m and it is subjected to a point load of 200kN from the left hand hinge at a distance of 6m.****Determine:**

- i) Magnitude of Resultant reactions and direction of Resultant reactions
- ii) Radial shear and normal thrust at quarter span from the left support